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ONLINE ACTIVITY #2
Personalized Linux, and Git Cheat Sheet

1. Basic Linux Commands

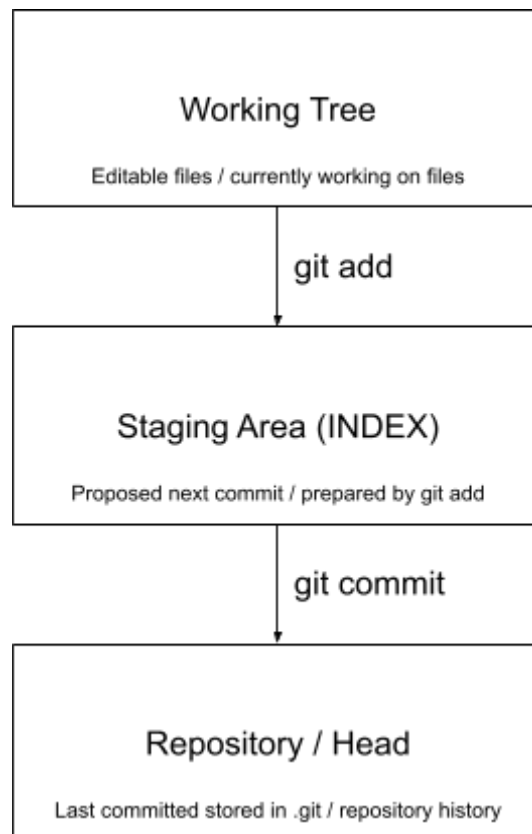
pwd Print the current working directory. Use pwd -P for the physical path.

Command	Sample Usage	Purpose / useful note
pwd	pwd	Print the current working directory. Use pwd -P for the physical path.
ls	ls -la	Lists directory contents. <code>-l</code> gives a detailed listing and <code>-a</code> includes entries whose names begin with <code>`.`</code> .
cd	cd Documents	Changes the shell's current directory. <code>cd ..</code> moves to the parent directory, while <code>cd -</code> uses the previous directory stored in <code>OLDPWD</code> .
mkdir	mkdir -p project/src	Creates directories. <code>-p</code> creates missing parent directories and does not report an error merely because an existing parent directory is already present.
touch	touch index.html	Changes file timestamps. If the named file does not exist, it is normally created as an empty file.
cp	cp -r src backup/	Copies files or directories. <code>-r</code> performs recursive directory copying.
mv	mv old.txt new.txt	Moves or renames files and directories.
rm	rm file.txt	Removes files. Recursive options such as <code>-r</code> can remove directory trees, so the command should be used carefully.
cat	cat notes.txt	Copies a file's contents to standard output, making it convenient for quickly viewing small text files.
clear	clear	Clears the terminal screen and, where supported, its scrollbar buffer.
chmod	chmod u+x script.sh	Changes file access permissions. In the example, <code>u+x</code> adds execute permission for the user who owns the file.
grep	grep -n "TODO" app.py	Searches input for matching lines. <code>-n</code> prefixes each match with its line number.
head	head -n 10 log.txt	Show the first lines of a file; <code>-n</code> sets the count.

2. Basic Git Commands

Command	Sample Usage	Description	Common options / flags
git init	git init -b main	Creates an empty Git repository or reinitializes an existing one. A normal initialization creates the .git directory that contains Git's internal repository data. The -b / --initial-branch option selects the initial branch name.	-b, --initial-branch, --bare
git status	git status -sb	Shows the state of the working tree and staging area. It reports differences between the index and HEAD, differences between the working tree and index, and untracked files that are not ignored.	-s / --short, -b / --branch, --porcelain
git add	git add -A	Copy selected working-tree content into the index for the next commit.	-A / --all; -p / --patch; -u / --update
git commit	git commit -m "Add README"	Record the current index as a new commit. -a auto-stages tracked modifications/deletions, not new untracked files.	-m; -a; --amend g
git log	git log --oneline --graph	Show commit history; output is reverse chronological by default.	--oneline; --graph; --all; --decorate
git diff	git diff --staged	Show changes. Plain git diff: working tree vs index; --staged: index vs HEAD.	--staged / --cached; --stat
git restore	git restore --staged file.txt	Restore paths. Without --staged, working-tree content comes from the index by default; with it, the index comes from HEAD by default.	--staged; --source=; -p
git branch	git branch -av	Create/list/manage branches; with no branch name it lists local branches.	-a / --all; -d; -m; -v
git checkout / git switch	git switch -c feature	Switch branches. switch -c creates and switches; checkout -b is the older equivalent. Checkout can also restore files.	switch -c; checkout -b; --detach
git merge	git merge feature	Integrate another branch/commit history into the current branch; conflicts may require manual resolution.	--no-ff; --ff-only; --abort; --continue
git tag	git tag -a v1.0 -m "Release"	Create/list/delete tags that name important points in repository history.	-a; -m; -l; -d
git clone	git clone --depth 1 URL	Copy an existing repository into a new directory and set up remote-tracking branches.	-b / --branch; --depth; --recurse-submodules

3. Three-Tree Architecture of GIT



References

1. Git Reference, individual command pages: <https://git-scm.com/docs>
2. Pro Git 2e, "Reset Demystified - The Three Trees":
<https://git-scm.com/book/en/v2/Git-Tools-Reset-Demystified>
3. GNU Coreutils Manual 9.11:
<https://www.gnu.org/software/coreutils/manual/coreutils.html>
4. GNU Bash Reference Manual (cd builtin):
<https://www.gnu.org/software/bash/manual/>
5. Linux man-pages, clear(1): <https://man7.org/linux/man-pages/man1/clear.1.html>
6. GNU Grep Manual 3.12: <https://www.gnu.org/software/grep/manual/grep.html>

AI USE DECLARATION

The author used ChatGPT (GPT-5.6 Sol) to research official Linux/Git documentation, cross-check command behavior and flags, organize the content, and prepare the cheat-sheet layout. The author reviewed and edited the content as needed and takes full responsibility for the final work.